MAE 184 Flight Simulation Techniques
Homework #2 Solution

Problem 2.1

Estimate the maximum airspeed for the jet aircraft if it is flying a constant altitude of 4.5 km.

mass, m = 12000 kg
wing reference area, S = 29 m^2

CD = CD0 + K*CL*CL
CD0 = 0.04 + 0.02*tanh[5.0*(M-0.95)]
K = 0.7 + 0.2*tanh(5.0*(M-0.95)]

Engine: (M = Mach number, h = altitude [km], T = max thrust [kN])
T = 78.7 + 13.3*M - 6.3*h + 0.13*h*h + 7.3*M*M – h*M

Standard Atmosphere at 4.5 km:
density, rho = 0.78 kg/m^3
speed of sound: sos = 322 m/s, M = V/sos

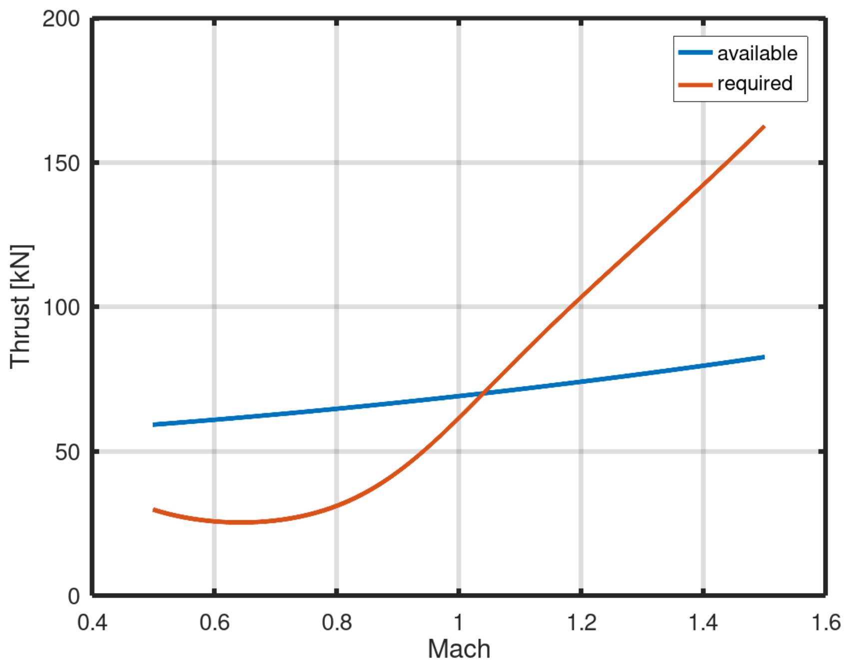
maximum airspeed = _________________ [m/s]

MATLAB/OCTAVE

```
% Problem 2.1

g = 9.8; % gravity [m/s2]
rho = 0.78; % density [kg/m3]
sos = 322; % speed of sound [m/s]
h = 4.5; % altitude [km]
S = 29.0; % wing reference area [m2]
mass = 12000; % aircraft mass [kg]

W = mass*g; % weight [N]

M = 0.5:0.01:1.5; % Mach numbers to test

for i=1:length(M)
    % thrust available [kN]
    TA(i) = 78.7 + 13.3*M(i) - 6.3*h + 0.13*h*h + 7.3*M(i)*M(i) - h*M(i);
    CD0 = 0.04 + 0.02*tanh(5.0*(M(i)-0.95));
    K = 0.7 + 0.2*tanh(5.0*(M(i)-0.95));
    V = M(i)*sos;
    % thrust required [kN]
    TR(i) = 0.001*(0.5*rho*V*V*S*CD0 + 2.0*K*W*W/rho/V/V/S);
end

plot(M,TA,"linewidth",3.0,M,TR,"linewidth",3.0)
set(gca,"linewidth",3.0,"fontsize",14);
xlabel("Mach","fontsize",16);
```

```
ylabel("Thrust [kN]", "fontsize",16);
legend("available", "required")
grid on;
```

ANSWER: Max Mach = 1.04, Max Airspeed = 335 m/s



Problem 2.2

The first flight of Blue Origin’s New Glenn rocket took place on 16 JAN 2025. The first stage of the rocket is equipped with seven (7) BE-4 motors. Each motor has a specific impulse of 340 seconds and produces 2,400 kN of thrust at sea level. The second stage is powered by two (2) BE-3U motors that each provide 770 kN of thrust with a specific impulse of 445 seconds. Simulate the rocket’s trajectory for 800 seconds after the initial time using a Runge-Kutta method and a step size of one (1) second. Plot the simulated vehicle airspeed [km/s] on the y-axis against time [s] on the x-axis. Compare your simulated airspeed by also plotting the reported airspeed trajectory. Be sure to label each axis and include a legend to mark each curve. Submit your plot using a standard image format (PNG, JPG, BMP).

$$\frac{dV}{dt}=\frac{T}{m}-g_0\left(\frac{R_E}{R_E+h}\right)^2\sin\gamma$$

$$\tau\frac{d\gamma}{dt}+\gamma=0$$

$$\frac{dh}{dt}=V\sin\gamma$$

$$\frac{dm}{dt}=-\frac{T}{g_0 I_{sp}}$$

RE = earth radius = 6378 km
g0 = sea-level gravity = 9.8 m/s^2
tau = guidance time constant = 120.0 sec

A. initial conditions:

time, t = 45 [sec]
airspeed, V = 45 [m/s]
flight path angle, gamma = 90 [deg]
altitude, h = 1600 [m]
mass, m = 1,200,000 [kg]

B. flight events:

(1) Main Engine Cut-Off (time = 180 sec)
  a. shut down first stage motors
  b. drop first stage structure, m = 240,000 [kg]
  c. start second stage motors
(2) Second-Stage Cut-Off (time = 780 sec)
  a. shut down second stage motors

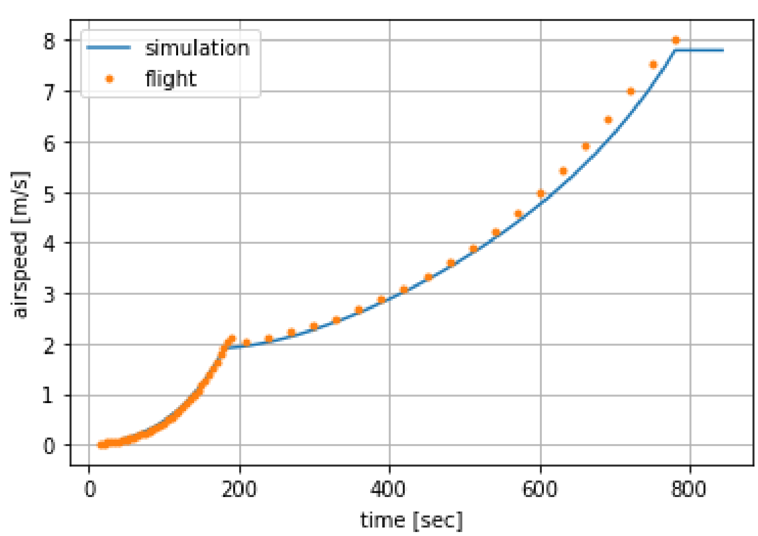
C. comparison data file (ng1.csv)

column #1 = time [sec]
column #2 = aispeeed [m/s]

PYTHON:

```
import numpy as np
import matplotlib.pyplot as plt

def EOM(s):
    """3DOF Equations of Motion."""

    g0 = 9.8           # sea-level gravity [m/s/s]
    Re = 6378000.0     # Earth radius [m]

    time = s[0]        # time [sec]
    V = s[1]           # airspeed [m/s]
    gam = s[2]         # vertical flight path angle [rad]
    alt = s[3]         # altitude [m]
    mass = s[4]        # mass [kg]
    Thrust = s[5]      # thrust [N]
    Isp = s[6]         # specific impulse [s]

    R = alt + Re      # radius from Earth center [m]

    sdot = np.zeros((7,1))

    # state rates

    sdot[0] = 1.0
    sdot[1] = Thrust/mass - g0*(Re/R)*(Re/R)*np.sin(gam)
    sdot[2] = -gam/120.0
    sdot[3] = V*np.sin(gam)
    sdot[4] = -1.0*abs(Thrust)/g0/Isp
    sdot[5] = 0.0
    sdot[6] = 0.0

    return sdot

def ModEuler(xold, h):
    """Modified Euler Integration."""
    r1 = EOM(xold)
    r2 = EOM(xold + h*r1)
    xnew = xold + 0.5*h*(r1+r2)
    return xnew

# simulation setup
dt = 1.0  # integration time step [s]
nsteps = 800 # number of time steps
xs = np.zeros((7,1,nsteps)) # storage

# set initial conditions
x = np.array([[45.0], # start time [s]
              [100.0], # airspeed [m/s]
              [0.5*np.pi], # vertical f.path [rad]
              [1600.0],  # altitude [m]
```

```
              [1200000.0],  # mass [kg]
              [7*2400*1000],  # thrust [N]
              [340.0]]) # specific impulse [s]

# run simulation
flight_phase = 0
for i in range(nsteps):

    if ((flight_phase == 0) and (x[0] > 180.0)):  # MECO
        x[5] = 0.0  # shut down first stage
        x[4] = x[4] - 240000  # drop booster mass
        x[5] = 2.0*770.0*1000.0  # ignite second stage motors
        x[6] = 445.0  # second stage Isp
        flight_phase = 1

    if ((flight_phase == 1) and (x[0] > 780.0)):  # SECO
        x[5] = 0.0  # shut down second stage
        flight_phase = 2

    xs[:,:,i] = x # store state vector

    x = ModEuler(x, dt) # Euler step

# simulation result
times = xs[0,0,:] # time vector [s]
vels = 0.001*xs[1,0,:] # airspeed [km/s]

# flight data
fdat = np.loadtxt("ng1.csv",delimiter=',')
timef = fdat[:,0]
velf = 0.001*fdat[:,1]

# plot output
plt.plot(times, vels, label = "simulation")
plt.plot(timef, velf, '.', label = "flight")
plt.xlabel('time [sec]')
plt.ylabel('airspeed [m/s]')
plt.grid()
plt.legend()
```



Problem 2.3

An airplane is beginning its final approach to an airport at an airspeed of 33 m/s. Air Traffic Control has requested the pilot to perform a “standard rate turn”; that is, to turn at a rate that would result in a 360 degree change in flight path heading (chi) in two minutes. What bank angle (mu) is required for this airplane to fly a standard rate turn at constant altitude?

a. 2.5 deg

b. 10 deg

c. 68 deg

d. 84 deg

MATALB/OCTAVE

```
V = 33.0; % airspeed [m/s]
g = 9.8; % gravity [m/s/s]
dchi_dt = (pi/180)*360.0/120; % heading rate [rad/s]

mu = (180/pi)*atan(V*dchi_dt/g)
```

ANSWER: bank angle, mu = 10 deg

Problem 2.4

An aircraft simulation uses quaternions to represent the relationship between the local and body-fixed reference frames. Find the four quaternion values that are needed to set the initial aircraft state so that the equivalent Euler angles are: roll attitude (phi) = -45 deg, pitch attitude (theta) = +10 deg, and heading (psi) = 90 deg.

a. (b0,bx,by,bz) = (-0.123,-0.123,-0.696,0.696)

b. (b0,bx,by,bz) = (0.674,0.213,0.327,0.627)

c. (b0,bx,by,bz) = (-0.528,-0.640,0.558,0.035)

d. (b0,bx,by,bz) = (0.627,-0.327,-0.213,0.674)

MATLAB/OCTAVE

```
phi = -45.0*pi/180.0;
theta = 10*pi/180.0;
psi = 90*pi/180.0;

p2 = 0.5*phi;
t2 = 0.5*theta;
s2 = 0.5*psi;

b0 = cos(p2)*cos(t2)*cos(s2) + sin(p2)*sin(t2)*sin(s2);
bx = sin(p2)*cos(t2)*cos(s2) - cos(p2)*sin(t2)*sin(s2);
```

```
by = cos(p2)*sin(t2)*cos(s2) + sin(p2)*cos(t2)*sin(s2);
bz = cos(p2)*cos(t2)*sin(s2) - sin(p2)*sin(t2)*cos(s2);

disp('b0 bx by bz')
disp([b0 bx by bz])
```

ANSWER: (b0,bx,by,bz) = (0.627,-0.327,-0.213,0.674)

Problem 2.5

Which set of differential equations can be used to simulated pure rolling motion (i.e. q = r = 0)?

ANSWER:

from lecture notes: $$\begin{bmatrix} \dot{b}_0 \\ \dot{b}_x \\ \dot{b}_y \\ \dot{b}_z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -b_x & -b_y & -b_z \\ b_0 & -b_z & b_y \\ b_z & b_0 & -b_x \\ -b_y & b_x & b_0 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

pure rolling motion: q = r = by = bz = 0

$$\begin{bmatrix} \dot{b}_0 \\ \dot{b}_x \\ \dot{b}_y \\ \dot{b}_z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -b_x & 0 & 0 \\ b_0 & 0 & 0 \\ 0 & b_0 & -b_x \\ 0 & b_x & b_0 \end{bmatrix} \begin{bmatrix} p \\ 0 \\ 0 \end{bmatrix}$$ yields: $$\begin{bmatrix} \dot{b}_0 \\ \dot{b}_x \\ \dot{b}_y \\ \dot{b}_z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -b_x p \\ b_0 p \\ 0 \\ 0 \end{bmatrix}$$

final answer: $\dot{b}_0 = -\frac{1}{2} b_x p$ and $\dot{b}_x = \frac{1}{2} b_0 p$ and $\dot{b}_y = \dot{b}_z = 0$